

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

9648/01

Paper 1 Multiple Choice

15 September 2016

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your Biology class, registration number and name above and on the Answer Sheet provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

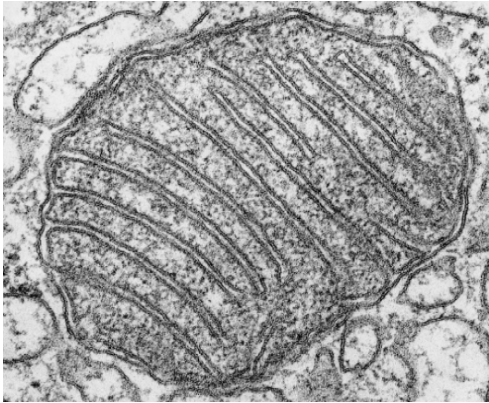
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **23** printed pages.

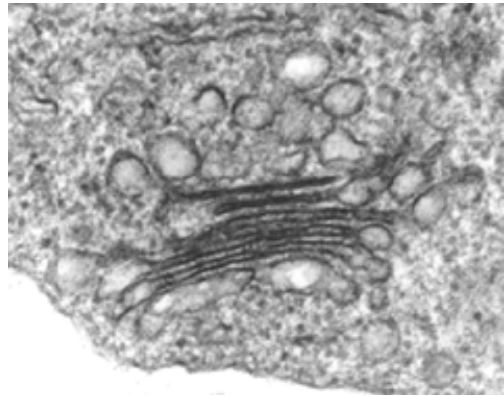
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- 1 The electron micrographs, which are taken at different magnifications, show four different organelles that can be found in different eukaryotic cells.

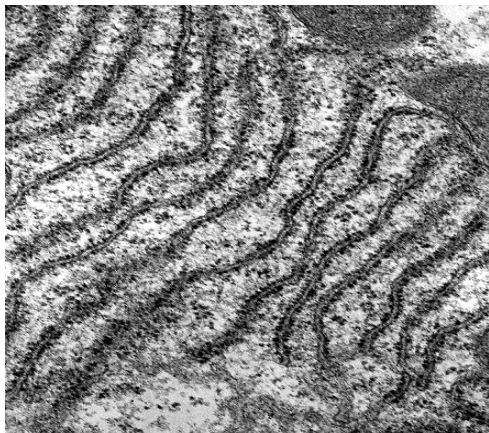
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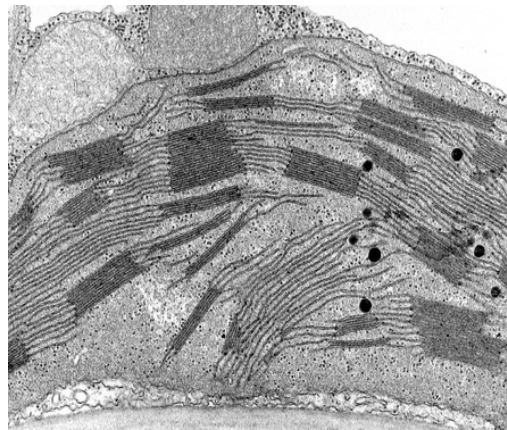
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3



4



Which organelle(s) contain(s) nucleic acids?

- A 4 only
- B 1 and 4 only
- C 1, 3 and 4 only
- D 1, 2, 3 and 4

2 Which statements about membrane fluidity are correct?

- 1 The less unsaturated the fatty acid chains of the phospholipids, the more fluid the membrane is.
- 2 The greater the amount of cholesterol in the membrane, the less fluid the membrane is at high temperatures.
- 3 The longer the hydrocarbon tails of the phospholipids, the more fluid the membrane is.
- 4 The lower the temperature, the less fluid the membrane is.

- A** 1 and 3
B 2 and 4
C 1, 2 and 3
D 2, 3 and 4

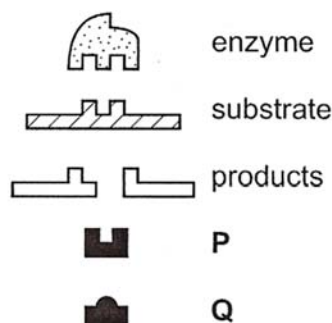
3 The following statements describe three orders of structure of the insulin molecule.

- 1 The molecule consists of two polypeptide chains joined and folded around one another.
- 2 The sequence and number of amino acids in each polypeptide chain are known.
- 3 The amino acids in each chain are coiled into a helix and held in position by hydrogen bonds.

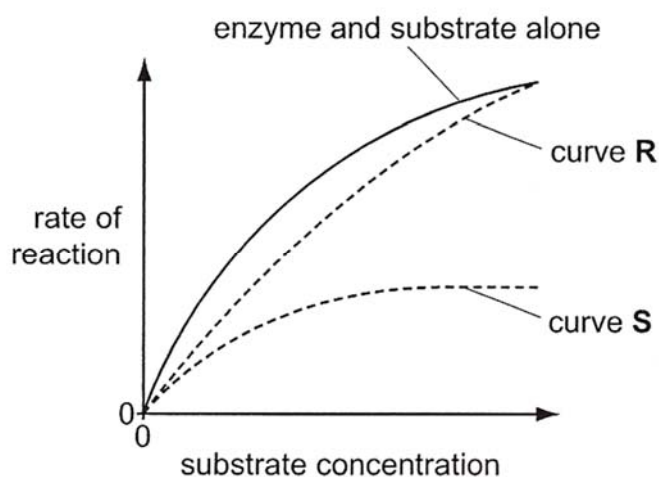
Which order is described by each statement?

	1	2	3
A	primary	secondary	tertiary
B	quaternary	primary	secondary
C	quaternary	primary	tertiary
D	secondary	tertiary	primary

- 4 The diagram shows an enzyme molecule with its normal substrate and products. P and Q are other molecules that can bind to the enzyme.



The graph shows the effect of P and Q on the rate of reaction of the enzyme at different substrate concentrations.



Which statement correctly describes the activity of the enzyme?

- A P is a competitive inhibitor that binds to the active site, resulting in curve R.
- B P is a non-competitive inhibitor that distorts the shape of the enzyme, resulting in curve S.
- C Q is a competitive inhibitor that distorts the shape of the enzyme, resulting in curve R.
- D Q is a non-competitive inhibitor that binds to the active site, resulting in curve S.

- 5 How does the second meiotic division differ from mitosis?
- A Chiasmata form between the chromatids of a bivalent in the second meiotic division but not in mitosis.
 - B Each chromosome replicates to form two chromatids during metaphase in the second meiotic division but not in mitosis.
 - C Exchange of genetic material occurs between chromatids in the second meiotic division but not in mitosis.
 - D The separating chromatids of a pair differ genetically in the second meiotic division but not in mitosis.
- 6 Suppose a cell with 14 chromosomes divides mitotically and one of the two new cells has 13 chromosomes and the other has 15 chromosomes.

At which phase of the cell cycle could an error have occurred and resulted in the unequal number of chromosomes in the two new cells?

- A anaphase
- B interphase
- C prophase
- D telophase

- 7 The following DNA sequence of the coding strand, which is complementary to the mRNA, is taken randomly from a bacterial genome.

3' TTACGCTTCGAAATAGGAATATCATAGGCT 5'

This DNA sequence is cloned into a plasmid, which is introduced into a suitable host.

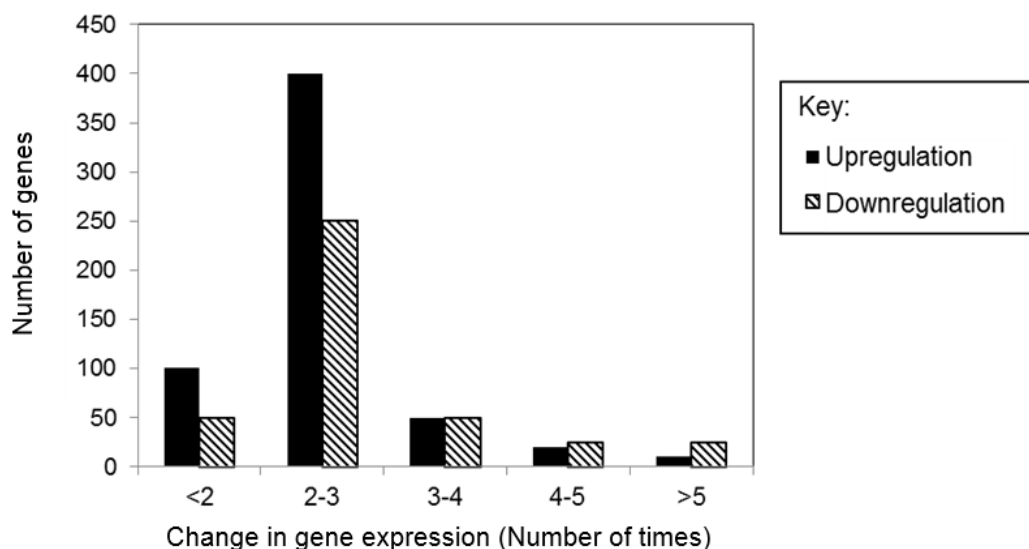
The table shows the mRNA codons for some amino acids.

arg	CGA, CGG, AGA, AGG	leu	CUU, CUC, CUA, CUG
asp	GAU, GAC	lys	AAA, AAG
ile	AUU, AUC, AUA	phe	UUU, UUC
met	AUG	ser	UCA, UCG, AGU, AGC
stop	UAG, UGA, UAA	tyr	UAU, UAC

What are the first four amino acids of the polypeptide generated from this DNA sequence?

- A met-arg-ser-lys
 B met-arg-ser-phe
 C met-ile-phe-leu
 D met-tyr-lys-asp
- 8 What is true about the regulation of gene expression in both prokaryotes and eukaryotes?
- 1 involves histone modifications
 - 2 involves helicase to separate the DNA so that transcription can take place
 - 3 involves binding of specific transcription factors to enhancers or silencers that are some distance from the gene(s) to be transcribed
 - 4 involves binding of a protein that can regulate transcription of several genes at the same time
- A 4 only
 B 2 and 3 only
 C 3 and 4 only
 D 1, 2 and 4 only

- 9 The graph shows the changes in gene expression in the liver cells of a group of mice after oxygen deprivation for four minutes.



What can result in all the observed changes?

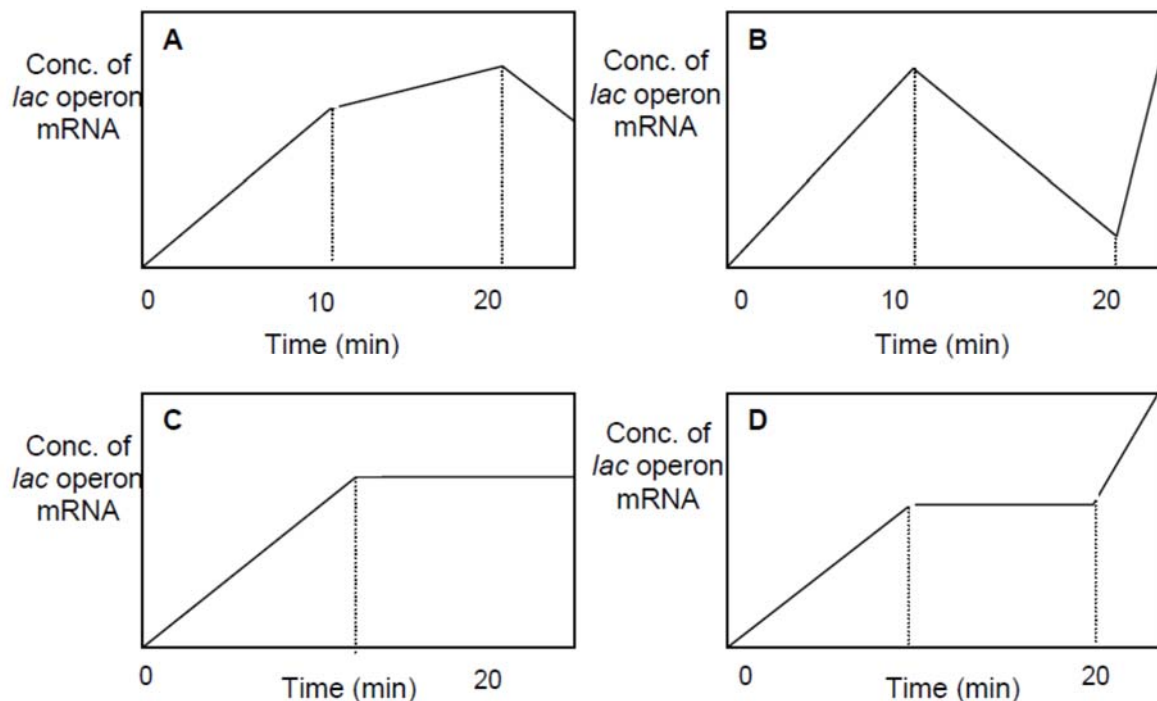
- A activation of specific transcription factors
 - B alternative splicing of mRNA
 - C increase in activity of histone acetyltransferase
 - D methylation of cytosine at the promoters of the genes
- 10 Which statement best defines an oncogene?
- A An oncogene codes for a cell cycle control protein.
 - B An oncogene codes for a mutated form of a protein that forms part of a signal transduction pathway.
 - C An oncogene codes for a protein that prevents the cell from undergoing apoptosis.
 - D An oncogene is a dominantly expressed mutated gene that gives a cell a growth advantage.

- 11** A scientist is studying a strain of bacteria whose genes are commonly transferred to other bacteria.

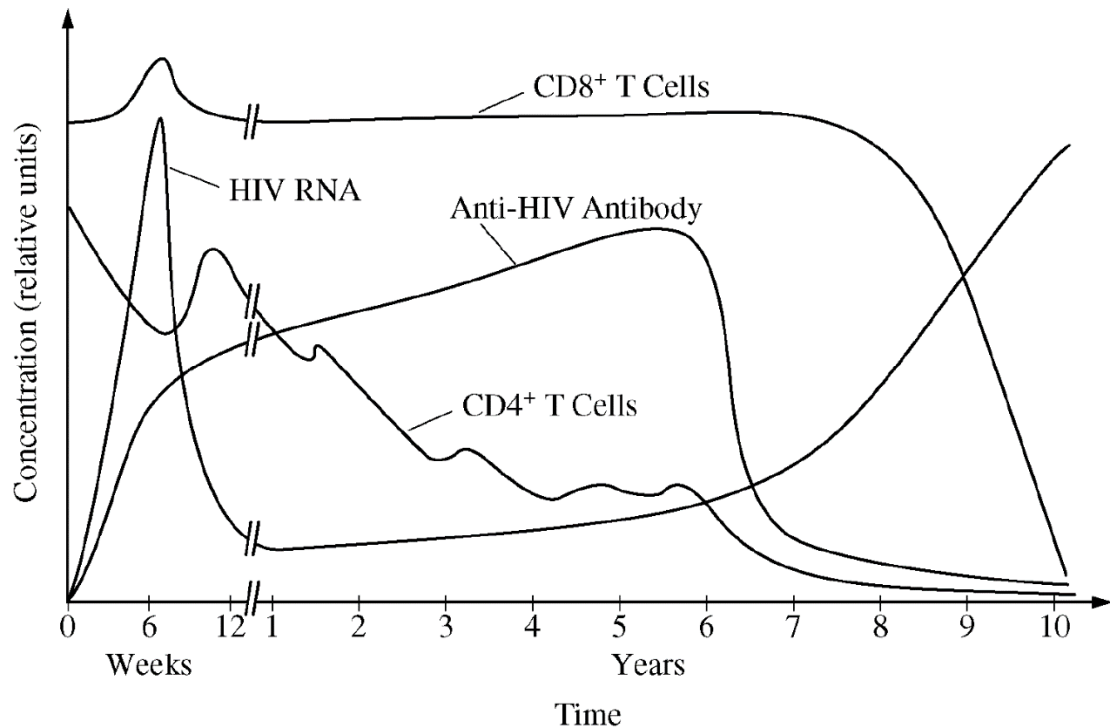
What could serve as evidence that the genes are transferred through specialized transduction?

- A** Cells that are treated with calcium ions show a higher rate of gene transfer.
B F plasmid is always transferred from donor cell to recipient cell.
C Only certain genes are transferred in the process.
D The strain of bacteria is often infected by a virulent phage.
- 12** IPTG is an analogue of lactose that binds to the *lac* repressor in the same fashion as allolactose. However, it cannot be metabolized by β -galactosidase. *E. coli* cells, which were grown in the absence of lactose and glucose, were initially supplemented with IPTG. After 10 minutes, glucose was added to the cells. 10 minutes later, cAMP was added to the cells.

Which graph best represents the amount of *lac* operon mRNA during the time course of the experiment?



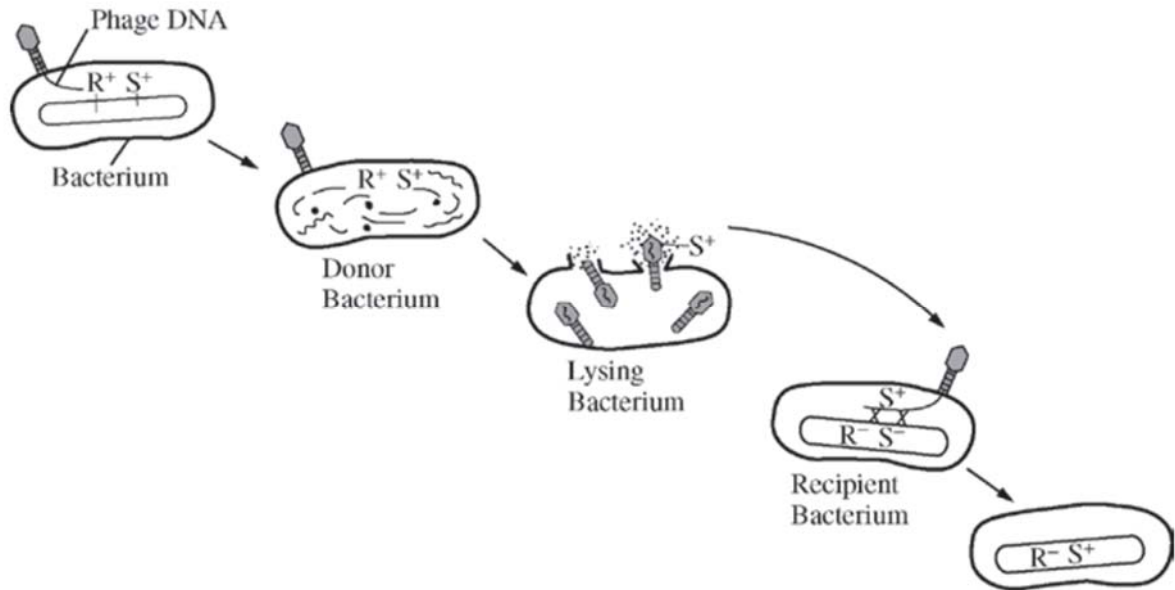
- 13 The graph shows the relationship among serum concentrations of human immunodeficiency virus (HIV) RNA, anti-HIV antibody, CD4⁺ and CD8⁺ T-cells over a ten-year period following infection of an individual by the virus.



What is the primary explanation for the drop in serum concentration of HIV-RNA during 6- to 12- week period following infection?

- A The viruses enter host cells via endocytosis to evade the immune system.
- B The viruses have a protein that accelerates viral replication.
- C The viruses that are actively replicating are eliminated by the immune system.
- D The viruses undergo rapid mutation.

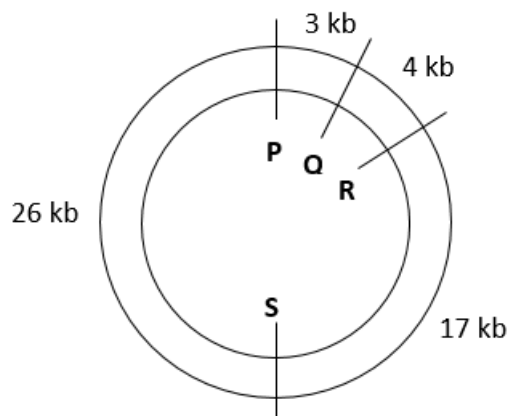
- 14 The diagram shows several steps in the process of bacteriophage transduction in bacteria.



Which statement explains how genetic variation in a population of bacteria results from this process?

- A Bacterial proteins transferred from the donor bacterium by the phage to the recipient bacterium recombine with genes on the recipient's chromosome.
- B DNA of the recipient bacterial chromosome undergoes recombination with DNA introduced by the phage from the donor bacterium, leading to a change in the recipient's genotype.
- C The phage infection of the recipient bacterium and the introduction of DNA carried by the phage cause increased random point mutations of the bacterial chromosome.
- D The recipient bacterium incorporates the transduced genetic material coding for phage proteins into its chromosome and synthesizes the corresponding proteins.

- 15 The diagram shows a plasmid with the positions of four restriction sites P to S indicated.



Copies of the plasmid were cut using two different restriction enzymes at a time, and the resulting fragments were separated by gel electrophoresis.

The diagram shows the results following gel electrophoresis of three samples.

Length of fragment (kb)	Restriction enzymes used		
	<i>BalI</i> and <i>EcoRI</i>	<i>PvuII</i> and <i>AvaI</i>	<i>PvuII</i> and <i>EcoRI</i>
40	■		
30		■	■
20		■	
10	■		■

Which row correctly matches the four restriction enzymes to their respective restriction sites on the plasmid?

	P	Q	R	S
A	<i>AvaI</i>	<i>EcoRI</i>	<i>PvuII</i>	<i>BalI</i>
B	<i>BalI</i>	<i>AvaI</i>	<i>EcoRI</i>	<i>PvuII</i>
C	<i>EcoRI</i>	<i>PvuII</i>	<i>BalI</i>	<i>AvaI</i>
D	<i>PvuII</i>	<i>BalI</i>	<i>AvaI</i>	<i>EcoRI</i>

16 The statements are about the preparation and application of DNA libraries.

- 1 A cDNA library allows the study of the functions of introns of specific genes.
- 2 A genomic library enables detection of genes that, in the host, have no detectable level of expression.
- 3 Alternative splicing can be studied using a cDNA library.
- 4 The preparation of a genomic DNA library requires restriction enzyme, reverse transcriptase and DNA ligase.

Which statements are correct?

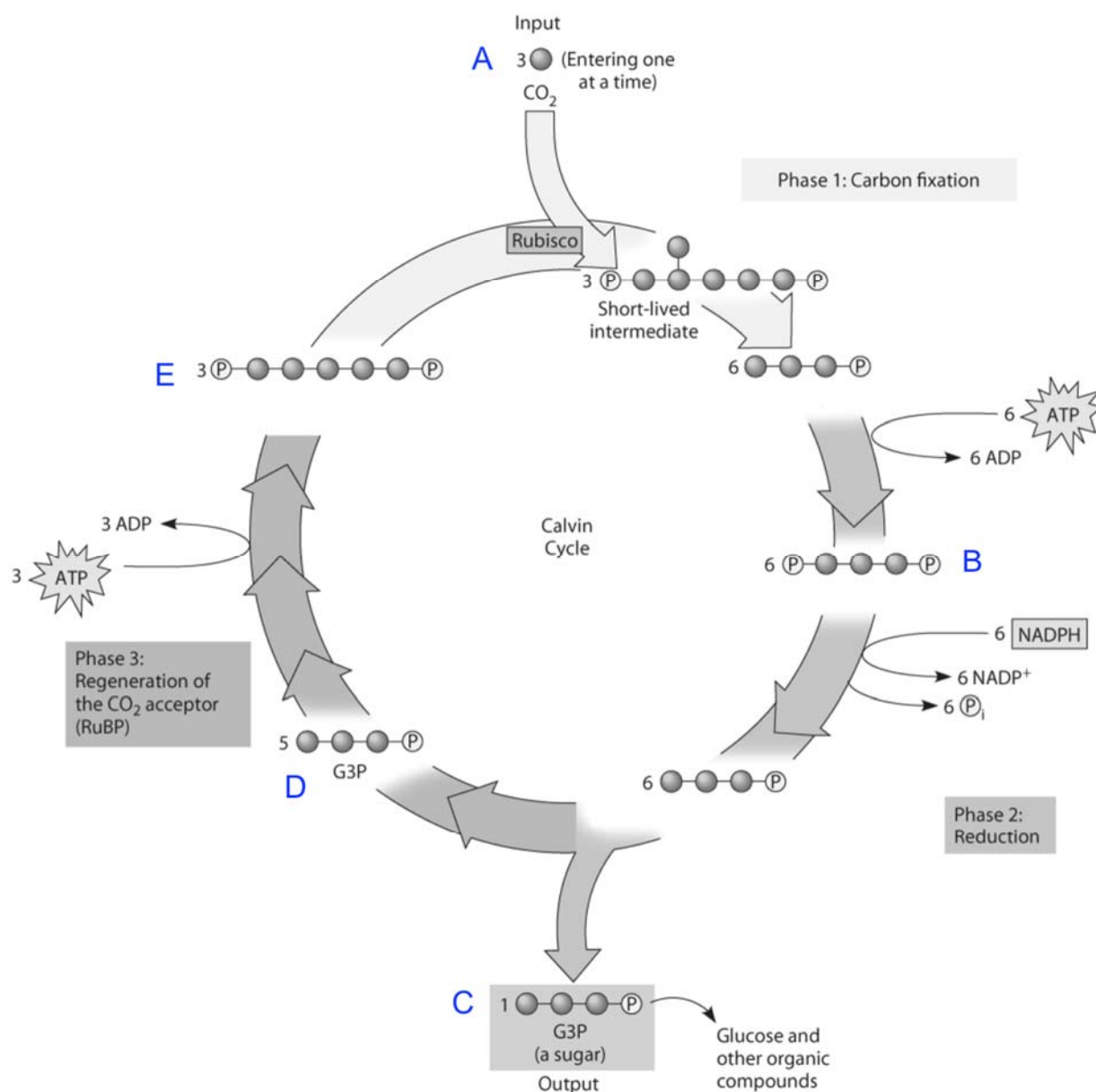
- A** 1 and 4 only
- B** 2 and 3 only
- C** 1, 2 and 4 only
- D** 2, 3 and 4 only

17 A sample of DNA was treated separately with two different restriction enzymes. The products of digestion were then run on a gel and stained with ethidium bromide.

Which of the following statements cannot account for the difference in the number of bands seen on the gel?

- 1 The two enzymes have active sites that are of different shapes.
 - 2 One of the enzymes produces blunt ends while the other produces sticky ends.
 - 3 Ethidium bromide binds with greater affinity to double stranded DNA than single stranded DNA.
- A** 1 only
 - B** 1 and 3 only
 - C** 2 and 3 only
 - D** 1, 2 and 3

18 The diagram shows some molecules that are involved in the Calvin cycle.



If ATP used by a plant is labelled with radioactive phosphorus, in which molecules would the radioactivity be measurable after one "turn" of the Calvin cycle?

- A A and B only
- B A, C and D only
- C B, D and E only
- D B, C, D and E only

- 19** Which statement about the Krebs cycle is correct?
- A** Oxygen is used to oxidise the acetyl group carbons of acetyl-CoA in the Krebs cycle.
 - B** Oxygen is not used in the Krebs cycle, so the cycle can occur in anaerobic conditions.
 - C** The Krebs cycle produces the water that is formed during the complete oxidation of glucose.
 - D** Three molecules of NADH and one molecule of FADH₂ are produced in one turn of the Krebs cycle.
- 20** Which statement correctly describes the role of lactate dehydrogenase?
- A** Lactate dehydrogenase catalyses the oxidation of pyruvate to lactate to regenerate NAD⁺.
 - B** Lactate dehydrogenase catalyses the reduction of pyruvate to lactate to regenerate NAD⁺.
 - C** Lactate dehydrogenase catalyses the oxidation of pyruvate to lactate to regenerate NADH.
 - D** Lactate dehydrogenase catalyses the reduction of pyruvate to lactate to regenerate NADH.
- 21** In humans, hair colour is determined by the pigment produced by hair follicle cells. The genes determining pigment colour are found on chromosome 15.

Which statement is true?

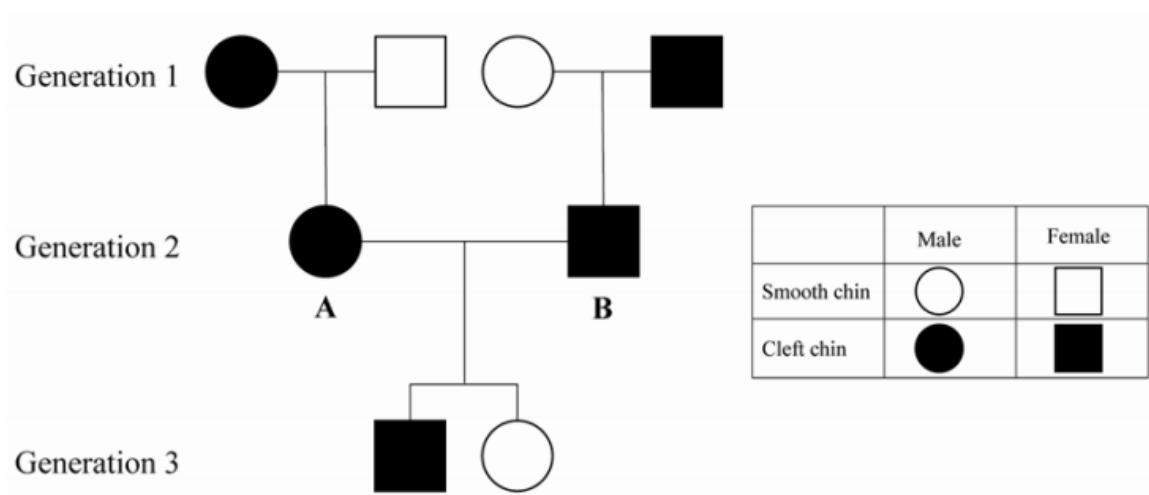
- A** Chromosome 15 in liver cells also contains genes determining pigment colour.
- B** Chromosome 15 in liver cells contains genes that are different from that of chromosome 15 in hair follicle cells as liver cells do not produce hair.
- C** Hair follicle cells do not contain homologous chromosomes of chromosome 15 as the cells only undergo mitosis.
- D** Homologous chromosomes of chromosome 15 in hair follicle cells contain the same genes but at different loci.

- 22** A strain of toad has only one nucleolus in the nucleus of each cell instead of the usual two. When such toads are mated, approximately one quarter of the offspring have two nucleoli per nucleus, one half have one nucleolus per nucleus, and one quarter have no nucleoli at all.

What is the most likely explanation of these results?

- A** The allele for the inheritance of two nucleoli per nucleus is dominant.
- B** The allele for the inheritance of two nucleoli per nucleus is recessive.
- C** The possession of one nucleolus per nucleus is due to the effect of crossing over.
- D** The possession of one nucleolus per nucleus is due to the heterozygous condition.

- 23** The pedigree diagram shows the chin types in a family.



Which statement correctly describes the cleft chin allele and the smooth chin allele?

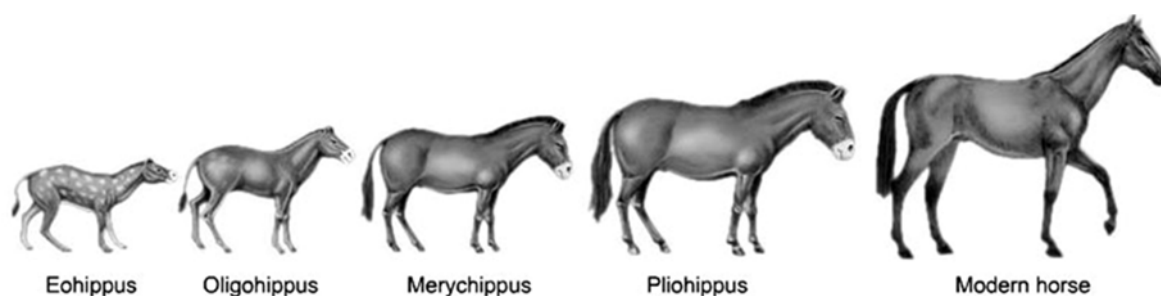
- A** Both alleles are codominant.
- B** Both alleles are linked on the X chromosome.
- C** The cleft chin allele is dominant over the smooth chin allele.
- D** The smooth chin allele is epistatic to the cleft chin allele.

- 24 The rate of divergence of different proteins is shown in the table below.

Protein	Amino acid substitutions per site per billion years
fibrinopeptide	8.3
lysozyme	2.0
α -globin	1.2
insulin	0.44
cytochrome c	0.3
histone H4	0.01

What may be concluded from the data about the neutral theory of molecular evolution?

- A All of these proteins are found in a common ancestor of all Earth's known living organisms before they started evolving at the molecular level.
- B Genetic drift could play a greater role in the evolution of the gene for fibrinopeptide resulting in greater number of amino acid substitutions per site per billion years.
- C Some essential proteins allow for fewer amino acid substitutions per site than others in order to maintain the same function.
- D The rate of neutral mutations that result in the amino acid substitutions for all proteins is the same.
- 25 The diagram suggests the evolution of horses beginning from the Eohippus 58 million years ago.



Fossil records show that the ancestor of the modern horse is believed to have had relatively short legs. According to Darwinian views, what best explains the evolution of horses?

- A acquired characteristics
- B directional selection
- C disruptive selection
- D stabilising selection

- 26** When organochlorine insecticides such as DDT were in widespread use, mosquitoes in malarial regions developed resistance more rapidly than houseflies in Britain.

What could account for the difference in the rate of development of resistance against organochlorine insecticides?

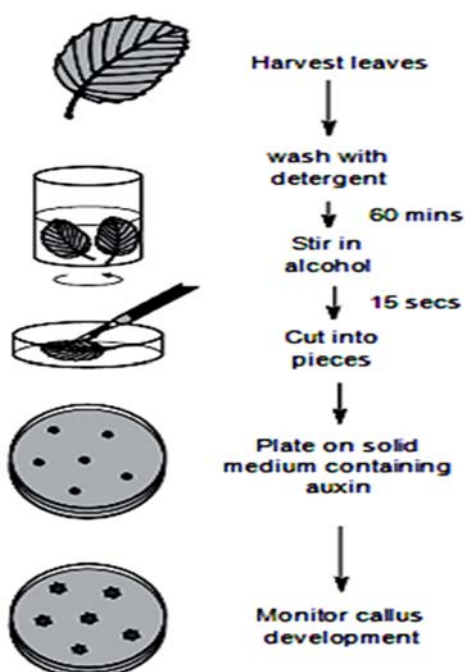
- A** More insecticides were used in Britain.
 - B** More insecticides were used in malarial regions.
 - C** Mosquitoes have fewer random mutations when exposed to insecticides.
 - D** Mosquitoes have more random mutations when exposed to insecticides.
- 27** Which process involves one stem cell giving rise to two distinct daughter cells: one copy of the original stem cell as well as a second daughter cell programmed to differentiate into a non-stem cell?
- A** asymmetric replication
 - B** differentiation
 - C** potency
 - D** self renewal
- 28** Which is not a source for stem cells?
- A** bone marrow
 - B** early embryos
 - C** egg cells
 - D** umbilical cord blood
- 29** Which is a reason why gene therapy involving the delivery of a normal allele of a proto-oncogene is not likely to be successful in the treatment of cancer?
- A** The proto-oncogene has many alleles controlling the production of the normal protein.
 - B** The delivery of the normal allele of the proto-oncogene is likely to cause the cell cycle to stop.
 - C** The normal allele for the proto-oncogene is not expressed in the presence of the dominant mutated allele.
 - D** The normal allele of the proto-oncogene codes for a protein that will allow the formation of a tumour.

- 30** The adenoviral vector is used to introduce the normal CFTR allele into patients with cystic fibrosis (CF). The main procedures involved in the treatment are listed below.
- Genes for replication are removed from the vector to prevent respiratory infections caused by adenoviruses.
 - Genes that causes the vector to elicit immune response are removed from the vector.
 - After treatment, patients are screened to ensure that replication of adenoviral vectors is not occurring in their system.
 - Patients are screened to ensure that shedding airway cells contain the normal CFTR allele

What limits the effectiveness of this type of gene therapy for CF?

- A** The adenoviral vector always causes infections in young and old patients.
 - B** The adenoviral vector delivers the normal CFTR allele to the airway cells that are gradually shed.
 - C** The adenoviral vector multiplies and make more copies of the normal CFTR allele.
 - D** The adenoviral vector will cause immune response in patients undergoing this treatment.
- 31** Which of the following is/are reason(s) for scientists to employ the method of plant cloning?
- 1 introduction of animal genes into plants
 - 2 create large changes at a rapid pace
 - 3 large scale production of pharmaceutical drugs
 - 4 selective breeding is too slow
- A** 3 only
 - B** 1, 3 and 4 only
 - C** 2, 3 and 4 only
 - D** 1, 2, 3 and 4

- 32 Callus tissue is produced using leaves by the technique shown in the diagram below. Using this, a student investigated the effect of two different concentrations of auxin on the volume of callus material produced. The table shows the results obtained over five days.



time (day)	volume of callus tissue (cm ³)	
	low concentration of auxin added	high concentration of auxin added
0	1.0	1.0
1	1.3	1.7
2	1.9	4.1
3	3.0	8.3
4	3.9	10.6
5	5.4	12.9

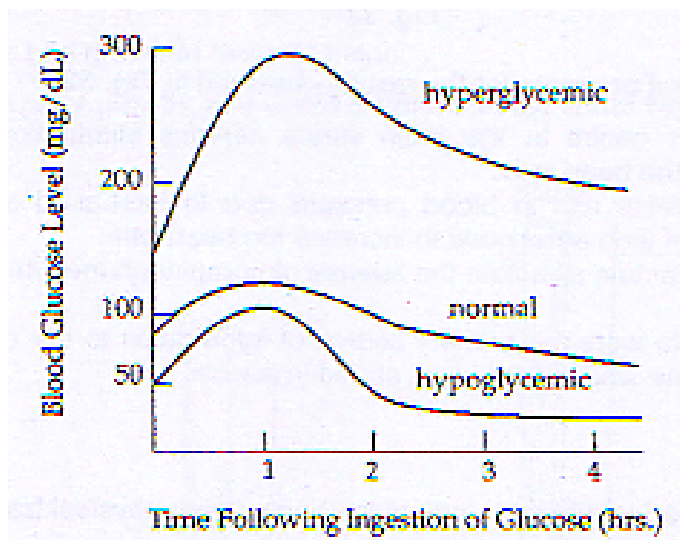
Which of the following statements about the experiment is / are true?

- 1 The leaves are disinfected and sterilised by washing with detergent and stirring in alcohol.
- 2 The leaves are wounded in order for callus tissue to form.
- 3 The concentration of auxin affects the volume of callus tissue formed.
- 4 Continued subculturing at three- to four-week intervals of small cell clusters taken from callus tissue can maintain the callus cultures for long periods of time.

- A 3 only
 B 1 and 3 only
 C 1, 2 and 3 only
 D 1, 2, 3 and 4

- 33** Which statements are possible issues of concern over the creation of genetically modified farmed animals?
- 1 Genetic engineering may result in the creation of new proteins that are harmful to the organisms that produce or consume them.
 - 2 Cross species gene transfer may compromise the genome integrity of the species involved.
 - 3 Over production of certain gene products may cause undue stress to the genetically modified farmed animals.
 - 4 Some genetically modified food products may not be acceptable to certain groups of people.
- A** 1 and 4 only
B 2 and 3 only
C 1, 3 and 4 only
D 1, 2, 3 and 4
- 34** Scientists are concerned about the escape of genetically modified mosquitoes into the wild. What is the most likely reason for this concern?
- A** The genetically modified mosquitoes may not survive in the wild.
B The mutation rate of the genetically modified mosquitoes will increase.
C The genetically modified mosquitoes may replace the wild mosquitoes population.
D The growth rate of the genetically modified mosquitoes will be affected.
- 35** Which statement best describes homeostasis?
- A** achieving independence from fluctuating internal conditions
B altering the external environment to accommodate the body's needs
C keeping the body in a fixed and unaltered state
D maintaining a near-constant internal environment

- 36 The graph shows the blood glucose levels for normal, hyperglycaemic and hypoglycaemic individuals. The values in the graph represent averaged data from 10 subjects for each condition.

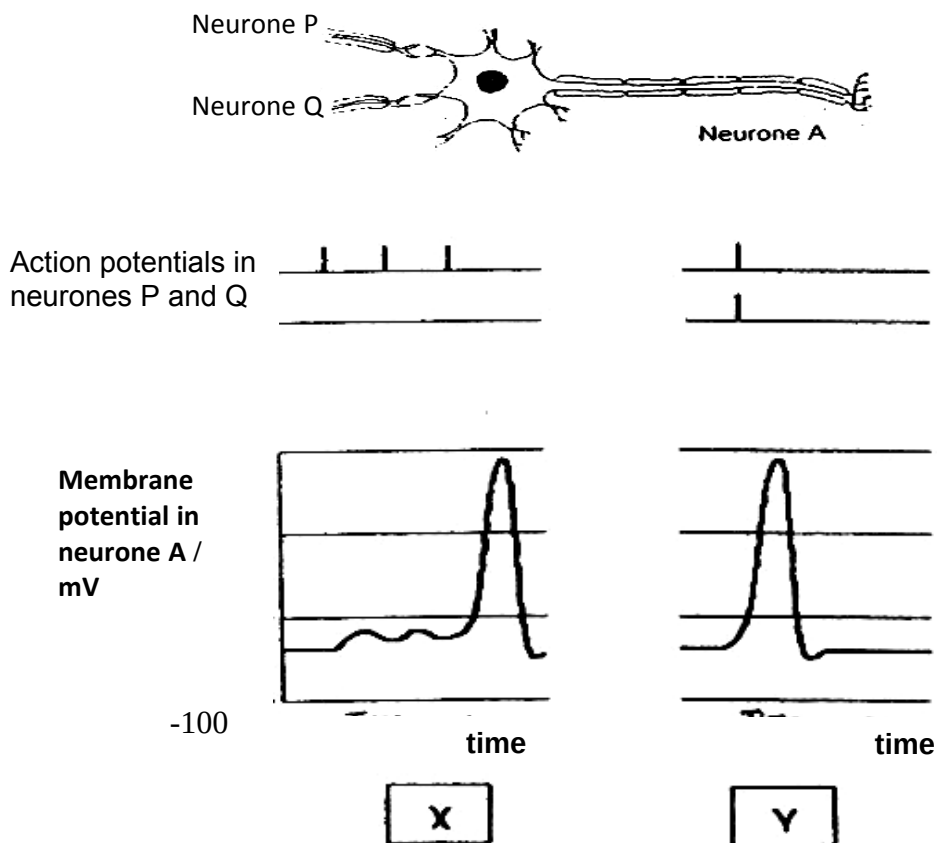


A glucose tolerance test was administered to evaluate the glucose levels in the blood of experimental subjects. Subjects were asked to fast for 12 hours before they were administered glucose at time 0 and then had blood samples drawn to assess the blood glucose levels. Glucose concentration was monitored every 30 minutes for a total of five hours. If the blood concentration exceeded 130 mg / dl, the subject was considered hyperglycaemic. Typically, hyperglycaemic patients exhibit symptoms such as thirst, increased appetite, and weight loss.

Why would a physician prescribe a diet consisting of several small meals to hypoglycaemic individuals?

- A to cause the liver to convert more glucose to glycogen
 - B to cause the pancreas to release more glucagon
 - C to maintain a steady blood glucose concentration
 - D to rapidly decrease the blood glucose concentration
- 37 What is the first event that happens to insulin receptor after binding to insulin?
- A Binding of insulin activates the receptor's cytosolic tyrosine kinase domain.
 - B Binding of insulin causes the receptors to dimerise.
 - C Binding of insulin causes the receptor to recruit a tyrosine kinase from the cytosol.
 - D Binding of insulin causes tyrosines in the receptor's C-terminal tail to become phosphorylated.

- 38 Which enzyme is activated by cyclic AMP in the signal transduction pathway after glucagon binds to its receptor?
- A adenylyl cyclase
B phosphorylase kinase
C protein kinase A
D protein kinase B
- 39 The diagram shows the effect of impulses from neurones P and Q on the production of an action potential in neurone A.



Which of the following best describes X and Y?

	X	Y
A	EPSP	IPSP
B	IPSP	EPSP
C	spatial summation	temporal summation
D	temporal summation	spatial summation

- 40** Certain drugs act at the synapses and affect the action of neurotransmitters.
The table shows the effects of four different drugs.

drug	effect
1	competes with acetylcholine at the receptor sites
2	inhibits the enzyme acetylcholinesterase
3	inhibits the opening of voltage-gated sodium ion channels
4	uncontrolled release of acetylcholine

Which combination of drugs will reduce the possibility of the formation of an excitatory postsynaptic potential?

- A** 1 and 3
- B** 2 and 4
- C** 1, 3 and 4
- D** 2, 3 and 4

- End of paper -